

Wheat (Gandum) Cultivation in Pakistan

A Complete, Practical Guide for Farmers — Punjab, Sindh, Khyber Pakhtunkhwa & Balochistan

Covering scientific background, agro-climatic requirements, recommended varieties, sowing, fertilizer, irrigation, pest and disease management, harvesting, storage, government policy, and modern & sustainable farming practices — specific to Punjab, Sindh, Khyber Pakhtunkhwa, and Balochistan.

Prepared as a practical, simple-English reference for farmers, students, extension workers, and agricultural chatbots.

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1. Introduction

Wheat, locally called gandum, is the most important food crop of Pakistan. It is grown almost everywhere in the country, from the irrigated plains of Punjab to the hilly tracts of Khyber Pakhtunkhwa and the dry lands of Balochistan. Wheat is sown in the Rabi (winter) season and harvested in spring. It provides the main source of calories for the vast majority of Pakistani households, mostly in the form of roti, naan, and chapati made from wheat flour (atta).

Because wheat touches food security, farm income, and national policy all at once, the government treats it as a strategic crop. Every year, federal and provincial governments announce sowing targets, support prices, and input subsidies to keep production high. For an ordinary farmer, growing wheat well means paying attention to variety choice, timely sowing, balanced fertilizer, proper irrigation, and pest and disease control.

This guide explains, step by step and in simple language, how wheat is grown in Pakistan today. It covers everything from land preparation to harvesting and storage, plus modern ideas like precision agriculture and sustainable farming, so that it can be used as a quick, reliable reference by farmers, students, extension workers, and by an AI chatbot answering farmer questions.

2. Scientific Name

The wheat grown across Pakistan belongs to the species *Triticum aestivum*, commonly known as bread wheat or common wheat. It belongs to the grass family, Poaceae (formerly called Gramineae). A small amount of durum wheat, *Triticum durum*, used for pasta and some local dishes, is also grown in limited areas, but bread wheat makes up almost all of the wheat area in the country.

Wheat is a self-pollinated, annual cereal crop with a fibrous root system and hollow, jointed stems called culms. The grain develops inside a spike (also called an ear or head), and each spike carries many spikelets, each holding two to five grains. Understanding this basic biology helps explain why the crop responds so strongly to sowing time, temperature, and nutrient supply during specific growth stages such as tillering, stem elongation, flowering, and grain filling.

3. Importance in Pakistan

Wheat is often called the backbone of Pakistan's agriculture and food security. It occupies more area than any other single crop in the country and provides a large share of the calories and protein in the average Pakistani diet. Per-person wheat consumption in Pakistan is among the highest in the world, so any shortfall in production directly affects flour prices and household budgets.

Economically, wheat contributes significantly to agricultural GDP and rural incomes, and it also has strong political importance because the government announces a minimum support price before every season to protect growers. A good wheat harvest generally means stable flour prices, lower import bills, and better rural cash flow that supports demand for other goods and services in the rural economy.

Wheat straw (bhoosa) is another valuable by-product, used as livestock fodder throughout the country, which links wheat farming closely with the livestock sector that many smallholder families depend on for extra income.

4. Major Producing Provinces and Districts

Punjab is by far the largest wheat-producing province, contributing roughly three-quarters of national wheat output. Leading Punjab districts include Bahawalnagar, Rahim Yar Khan, Bahawalpur, Jhang, Faisalabad, Layyah, Khanewal, Muzaffargarh, Dera Ghazi Khan, Sheikhpura, Rajanpur, Gujranwala, Vehari, Multan, Hafizabad, Lodhran, Sargodha, Mianwali, Okara, and Sialkot. A relatively small group of about twenty districts in Punjab alone produces more than half of the country's total wheat.

Sindh is the second-largest producer, with important wheat areas in Larkana, Shaheed Benazirabad (Nawabshah), Sukkur, Ghotki, Khairpur, Sanghar, Badin, and Thatta. Wheat in Sindh benefits from canal irrigation from the Indus River and, in recent seasons, from residual soil moisture left after floods.

Khyber Pakhtunkhwa grows wheat mainly in Peshawar, Mardan, Swabi, Charsadda, Bannu, Dera Ismail Khan, and the wider Malakand and Hazara divisions, using a mix of irrigated and rain-fed (barani) fields. Balochistan produces the smallest share nationally, but wheat is important locally in Quetta, Pishin, Mastung, Nasirabad, Jaffarabad, and Naseerabad division, where both irrigated and dryland sowing are practiced, often starting earlier than in the other three provinces.

5. Climate Requirements

Wheat is a cool-season crop that needs a moderate climate for good growth and grain filling. It grows best when average temperatures during vegetative growth stay between about 15 and 20 degrees Celsius, while cooler temperatures near 10 to 15 degrees Celsius favor good tillering. A germinating wheat seed needs soil temperature around 20 to 25 degrees Celsius for quick, uniform emergence.

The most sensitive period is grain filling in spring. If temperatures rise sharply above the optimum during this stage, especially with hot, dry winds (locally feared as 'loo' conditions arriving early), grains shrivel and yield drops quickly. Research shows that even one extra degree of heat above the ideal level during grain filling can cut yield by five to seven percent, which is why late sowing is discouraged across Pakistan.

Wheat also needs a relatively dry atmosphere at maturity to allow proper ripening and easy harvesting; excess humidity or untimely rain near harvest can cause lodging (crop falling over) and disease outbreaks. Overall, total crop water requirement across the season is roughly 270 to 515 millimetres, most of which in Pakistan is supplied through irrigation rather than rainfall alone.

6. Soil Requirements

Wheat can be grown on a wide range of soils, but it performs best on well-drained loam to clay-loam soils with good water-holding capacity. Very sandy soils drain and dry too quickly and need more frequent irrigation and fertilizer splits, while heavy, poorly drained clay soils can cause waterlogging and poor root development if not managed carefully.

The ideal soil pH for wheat is close to neutral, roughly 6.0 to 7.5. Much of Pakistan's irrigated land tends toward alkaline conditions with pH above 7.5 to 8.5 and can also suffer from low organic matter, so gypsum application and organic matter addition are commonly recommended on problem soils to correct pH and improve structure.

Because Pakistani soils are generally low in organic matter, farmers are increasingly advised to use farmyard manure (FYM), compost, or green manuring in rotation with wheat, and in some cases soil conditioners like potassium humate, to boost microbial activity, root development, and nutrient availability, particularly phosphorus, which becomes fixed and less available in alkaline soils.

7. Land Preparation

Good land preparation gives wheat a fine, firm, and weed-free seedbed, which is essential for even germination and strong early growth. After harvesting the previous crop (commonly cotton, rice, sugarcane, or maize), the field is usually given one deep ploughing followed by two to three cultivations with a tiller or disc harrow, each followed by planking (suhaga) to break clods and level the surface.

Laser land levelling has become a widely promoted practice in Punjab and Sindh because it creates a perfectly level field, which allows uniform water distribution, saves irrigation water, improves germination, and increases yield compared to traditionally levelled fields. Many provincial governments and donor projects subsidize laser levelling services for smallholders.

Where wheat follows rice in the same field (the rice-wheat cropping system common in central Punjab), stubble and straw need to be managed properly, either incorporated into the soil, used as fodder, or removed, rather than burned, since burning destroys organic matter and pollutes the air. Zero tillage and happy-seeder drills are gaining popularity here because they allow wheat to be sown directly into rice residue, saving time, fuel, and moisture.

8. Recommended Varieties (Pakistan)

Pakistan's national and provincial research institutes, especially the Wheat Research Institute (WRI) at Ayub Agricultural Research Institute, Faisalabad, and the National Agricultural Research Centre (NARC), Islamabad, regularly release new, higher-yielding, disease-resistant varieties, so farmers are advised to check the latest recommendations from their local agriculture extension office each season.

In Punjab, popular and recommended varieties include Akbar-2019, Dilkash-2020, Arooj-2022, Subhani-2021, Ghazi-2019, Nawab-2021, Anaj-2017, Bhakkar-2020, Galaxy-2013, and Ujala-2016, along with barani (rain-fed) varieties like Arooj and Barani-17 for the drier tracts of Chakwal, Attock, and the Pothohar plateau. These varieties combine good yield potential with resistance to yellow and brown rust, which are common threats in Punjab.

In Sindh, leading varieties include Sunehri, NIA Amber, TD-1, Khirman, Imdad-2005, Sassui, Sarang, and Benazir-2013, chosen for tolerance to the warmer Sindh climate and shorter growing season. In Khyber Pakhtunkhwa, Pirsabak, Siran, and other locally adapted lines are common, while in Balochistan, Zardana, Sariab-92, and other heat- and drought-tolerant lines suit the drier and more variable rainfall conditions. Farmers should always use certified seed from a trusted source rather than saving grain from the previous crop for too many years, because certified seed keeps disease resistance and yield potential intact.

9. Seed Rate

The standard seed rate recommended for irrigated wheat sown at the right time with a seed drill is about 50 kilograms per acre (roughly 125 kilograms per hectare). This gives an even plant population without excessive competition between plants for light, water, and nutrients.

When sowing is delayed beyond the ideal window, the seed rate should be increased to around 60 to 70 kilograms per acre, since late-sown wheat tillers less and needs more plants per unit area to compensate for the shorter growing season. Similarly, broadcast sowing (chatta method) generally requires a slightly higher seed rate than drill sowing because seed distribution is less uniform and germination is somewhat lower.

For barani (rain-fed) areas and for larger, bolder-seeded varieties, farmers may adjust the rate slightly upward, and it is always good practice to do a germination test on the seed lot before sowing so the actual seeding rate can be corrected for germination percentage rather than assuming 100 percent viability.

10. Seed Treatment

Treating seed before sowing is a simple, low-cost step that protects the crop from several early diseases and pests, especially loose smut, bunt, and seedling blight, as well as termite attack in areas where termites are a known problem. Seed treatment is strongly promoted by extension departments as one of the most cost-effective inputs in wheat production.

Common practice is to treat seed with a recommended fungicide, such as those containing carboxin, carbendazim, or triazole-based active ingredients, applied at the manufacturer's recommended dose per 50 kilograms of seed, mixed thoroughly so every seed gets a light, even coating before it dries and is sown. Where termites are a known risk, an appropriate insecticide seed dressing is added along with the fungicide.

Farmers should always wear gloves and a mask while treating seed, treat seed in a shaded area away from children and animals, and never use treated seed for food or feed purposes. Locally produced biofertilizer or Rhizobium-type seed inoculants are not typically needed for wheat since it is not a legume, but phosphate-solubilizing bacterial inoculants are increasingly promoted to improve phosphorus uptake in alkaline soils.

11. Sowing Time (Province-wise)

Sowing time is the single biggest factor influencing wheat yield in Pakistan, since even a few weeks of delay pushes grain filling into hotter spring weather and lowers yield sharply. Each province has its own recommended window based on local temperature patterns.

In Punjab, the ideal sowing window is from around the last week of October to 20 November for irrigated areas, with 1 to 10 November considered the peak optimum period for most varieties; barani (rain-fed) areas of the Pothohar plateau often start slightly earlier, from mid-October. In Sindh, sowing is generally done a little later, from around 1 November to the end of November, because of the warmer autumn climate.

In Khyber Pakhtunkhwa, the recommended window runs from about 15 October to 15 November, with cooler upland areas of Malakand and Hazara sowing earlier than the warmer valley floors near

Peshawar and Mardan. In Balochistan, sowing begins earlier than anywhere else in the country: upland areas around Quetta can start planting from around 1 October, while sowing in the plains and lower valleys extends from 1 November until as late as mid-December to February in some dryland tracts, reflecting the province's wide range of altitudes and climates.

12. Sowing Methods

Drill sowing with a seed-cum-fertilizer drill is the most recommended method across Pakistan because it places seed at a uniform depth (about 4 to 5 centimetres) and in straight rows spaced roughly 20 to 22.5 centimetres apart, giving even emergence, easier weeding, and better use of fertilizer placed near the root zone.

Broadcasting (chatta method), where seed is scattered by hand over a prepared or standing-water field and then covered by planking, is still common among smallholders, especially in parts of Sindh and rain-fed Punjab, but it uses more seed, gives uneven plant stands, and makes inter-row cultivation for weeds difficult.

Bed and furrow (ridge) sowing is being promoted, particularly in Punjab and Sindh, because it saves irrigation water, improves drainage in heavier soils, and allows easier movement through the field for weeding, spraying, and later intercropping. Zero tillage or happy-seeder sowing, where wheat seed is drilled directly into standing rice or cotton stubble without conventional ploughing, is expanding quickly in the rice-wheat belt of central Punjab because it saves time, diesel, and soil moisture between the two crops.

13. Fertilizer Recommendations (NPK)

General fertilizer recommendations for irrigated wheat in Pakistan are approximately 120 to 150 kilograms of nitrogen (N), 60 to 90 kilograms of phosphorus as P₂O₅, and 60 kilograms of potassium as K₂O per hectare, though the exact figures vary by soil test, province, and variety; extension departments always encourage soil testing before deciding the final dose.

In common farm units, this typically translates into roughly two bags of Urea (about 100 to 110 kilograms) plus one to one and a half bags of DAP (Di-Ammonium Phosphate) per acre, with potash (Sulphate of Potash or Muriate of Potash) added on soils known to be potassium-deficient, particularly sandy soils and some rice-wheat soils in central Punjab.

The recommended practice is to apply all of the phosphorus and potassium, plus about one-third to one-half of the nitrogen, at the time of sowing, placed close to the seed row using a drill. The remaining nitrogen is split into two top-dressings: one at the first irrigation around 20 to 25 days after sowing during the tillering stage, and the second at the second irrigation around 45 to 50 days after sowing during stem elongation. This split nitrogen approach avoids losses from leaching and volatilization and matches nutrient supply with crop demand at its most active growth stages. Zinc sulphate application, especially on rice-wheat rotation soils, is also commonly recommended where zinc deficiency has been confirmed.

14. Irrigation Schedule

Wheat is mostly grown under irrigation in Pakistan, since about 82 to 85 percent of the cultivated area relies on canal or tube-well water rather than rainfall alone. A well-managed irrigation schedule usually needs four to six waterings across the season, depending on soil type, rainfall, and variety duration.

The first irrigation (called 'first watering' or rauni/first irrigation after sowing) is critical and is normally applied 20 to 25 days after sowing, coinciding with the tillering stage, when the crop is establishing the tillers that will later carry grain-bearing spikes. The second irrigation follows at around 45 to 50 days after sowing, during stem elongation (jointing), and is equally important for building yield potential.

Further irrigations are typically applied at the booting stage, at flowering, and during grain-filling (milk to dough stage), spaced roughly three to four weeks apart depending on weather and soil moisture. Farmers are advised to avoid irrigation right before a rain event or during very hot, windy periods when lodging risk is high, and to stop irrigation about 10 to 15 days before the expected harvest so the crop dries down properly for a clean harvest and safe storage. Laser-levelled fields and bed-and-furrow sowing both help reduce the amount of water needed per irrigation while maintaining good yield.

15. Weed Management

Weeds compete strongly with wheat for water, nutrients, and light, and can reduce yield by 20 to 40 percent or more if left uncontrolled, especially in the first 30 to 45 days after sowing, which is the critical weed-competition period. The most troublesome weeds in Pakistani wheat fields include wild oats (Jangli Jai), canary grass (*Phalaris minor* / Baisan), *Chenopodium* (Bathu), *Convolvulus* (Lehli), and broadleaf weeds like *Malva* (Sonchara) and *Rumex* (Jangli Palak).

Cultural control starts with clean, certified seed free of weed seed contamination, proper crop rotation, and timely sowing, since early, vigorous wheat competes better with weeds than late-sown, weak crops. Manual or mechanical weeding (hand-pulling or hoeing) is still practiced on small farms, especially for scattered broadleaf weeds, but it is labour-intensive for larger holdings.

Chemical control with selective herbicides is now standard on most medium and large farms. Grassy weeds like wild oats and canary grass are controlled with herbicides based on active ingredients such as clodinafop, fenoxaprop, or pinoxaden, while broadleaf weeds are controlled with products containing 2,4-D, metsulfuron, or carfentrazone, usually applied 30 to 35 days after sowing when weeds are small and the crop has established well. Farmers should always follow label rates and timing, and rotate herbicide groups across seasons to slow the development of herbicide-resistant weed populations, which are an increasing problem in intensively cropped areas like central Punjab.

16. Insect Pests

The most economically important insect pest of wheat in Pakistan is the aphid, which sucks sap from leaves and stems and can also transmit viral diseases; heavy infestations, especially during mild, humid winters, reduce plant vigour and grain filling. Aphid populations are usually monitored from the tillering stage onward and controlled with recommended insecticides only when they cross the economic threshold, to protect natural predators like ladybird beetles.

Termites are a serious problem in sandy soils and rain-fed areas, attacking roots and the base of the stem, causing plants to dry up in patches (locally seen as 'white heads' or dead tillers); seed treatment

with insecticide and, where needed, soil application around germination, are the main control measures. Armyworms and cutworms occasionally cause damage by cutting seedlings at the base or defoliating young plants, especially after mild, wet weather.

Other occasional pests include termites' close relative the wireworm, storage weevils that attack harvested grain, and, in some seasons, locust swarms that can cause widespread damage if not controlled early through coordinated provincial and federal plant protection efforts. Rodents (field rats) also damage standing crop and stored grain in many districts and are controlled with approved rodenticides and field sanitation. Integrated Pest Management (IPM), which combines resistant varieties, timely sowing, field sanitation, and need-based rather than routine pesticide spraying, is the approach recommended by agriculture departments to keep pest control both effective and cost-efficient.

17. Diseases

Rust diseases are the biggest disease threat to wheat in Pakistan, appearing as yellow, brown, or black powdery pustules on leaves and stems. Yellow (stripe) rust is especially damaging in cooler, humid tracts of upper Punjab and Khyber Pakhtunkhwa, brown (leaf) rust is more common in central and southern Punjab and Sindh, and black (stem) rust can appear later in the season under warm, humid conditions. Growing resistant, recommended varieties is the first line of defence, backed up by fungicide sprays such as those containing propiconazole or tebuconazole when rust appears early and conditions favour its spread.

Smuts and bunts, including loose smut and Karnal bunt, replace healthy grain with black, foul-smelling spore masses and are largely controlled through certified, disease-free seed and proper seed treatment with fungicide before sowing, since these diseases are mainly seed-borne. Powdery mildew, a white, powdery fungal growth on leaves, can appear in cool, humid, densely sown fields, especially where excessive nitrogen has encouraged very thick crop canopies.

In recent years, wheat blast, caused by the fungus *Magnaporthe oryzae* pathotype *Triticum*, has emerged as a newer threat in some warmer, humid districts, causing bleached spikes and empty grain; awareness campaigns now urge farmers to report suspicious symptoms quickly to extension staff. Fusarium head blight and root/crown rot are also monitored, particularly where wheat follows maize or rice residue is left on the surface without proper decomposition.

18. Deficiency Symptoms

Nitrogen deficiency shows up as overall pale yellow-green colour, starting from older, lower leaves, along with stunted growth, thin stems, and poor tillering; it is common on sandy soils or where nitrogen has not been split properly across irrigations. Phosphorus deficiency causes a dull, dark green to purplish tint on leaves, especially in young plants during cold weather, along with poor root development and delayed maturity, and is common on alkaline, calcareous soils typical of much of Punjab and Sindh where phosphorus becomes chemically fixed.

Potassium deficiency appears as yellowing and browning (scorching) of leaf margins and tips on older leaves, weaker stems that are more prone to lodging, and poor grain filling; it is more likely on light, sandy soils and in intensive rice-wheat rotations where potassium removal is high. Zinc deficiency, a

widespread problem in Pakistan's calcareous, alkaline soils, causes interveinal chlorosis (yellow-white streaking between leaf veins), stunted growth, and delayed maturity, and is usually corrected with zinc sulphate applied to the soil or as a foliar spray.

Iron deficiency, less common but seen on very calcareous or waterlogged soils, produces distinct yellowing between the veins of young leaves while the veins themselves stay green; it is generally treated with foliar sprays of iron sulphate or chelated iron since soil-applied iron is quickly fixed in alkaline conditions. A basic soil test, done before sowing, remains the most reliable way to catch and correct nutrient problems before they become visible in the crop.

19. Harvesting

Wheat is ready for harvest when the crop turns golden-yellow, the straw becomes dry and brittle, and grain moisture drops to roughly 12 to 14 percent, which usually happens 4 to 5 months after sowing depending on variety and sowing date. In Sindh, harvesting typically starts around mid-March, moving into Punjab from late March through May, with Khyber Pakhtunkhwa and Balochistan's higher, cooler tracts harvesting later, sometimes into June.

Timely harvesting is important because delayed harvest exposes the crop to shattering losses (grain falling to the ground), lodging from wind or rain, and bird and rodent damage. Harvest can be done manually with a sickle, which is still common on very small holdings and in areas with cheap labour, or mechanically using a combine harvester, which cuts, threshes, and cleans the grain in one pass and is now widely used across Punjab and Sindh, especially on medium and large farms, because it saves time and reduces harvest losses.

Where manual harvesting is used, the cut crop is left to dry in the field for a few days before threshing with a mechanical thresher; care should be taken to thresh at the correct cylinder speed to avoid cracking grain or leaving unthreshed heads. Straw (bhoosa) should be collected and stored properly since it is a valuable livestock feed resource across rural Pakistan.

20. Storage

Proper storage protects wheat from insect pests, rodents, moisture damage, and fungal growth, all of which can cause serious quality and quantity losses after harvest. Before storage, grain should be dried to a safe moisture level of around 10 to 12 percent, since higher moisture encourages mould, heating, and insect activity, particularly of storage pests like the rice weevil, red flour beetle, and khapra beetle.

At the household and small-farm level, wheat is commonly stored in clean, dry, well-ventilated mud bins (bharoli/kothi), metal drums, or jute/polypropylene bags stacked on wooden pallets off the floor to avoid ground moisture, away from walls and with regular inspection for insect activity. Government-managed storage, through the Pakistan Agricultural Storage and Services Corporation (PASSCO) and provincial food departments, uses larger warehouses, silos, and covered plinths for procured wheat.

Recommended storage practices include cleaning storage structures before filling, using approved grain protectant insecticides or safe fumigation where necessary and legally permitted, avoiding mixing old and new stock, and checking stored grain periodically for temperature, moisture, and insect signs. Hermetic (airtight) storage bags and silos are increasingly promoted as a low-chemical alternative that

starves insects of oxygen and keeps grain quality high for longer periods, an approach particularly useful for smallholders who cannot afford large warehouse infrastructure.

21. Expected Yield in Pakistan

National average wheat yield in Pakistan has been around 2.9 to 3.2 tonnes per hectare in recent years, equivalent to roughly 28 to 32 maunds per acre on average across all types of farms, though this masks a wide range between well-managed and poorly managed fields. On a good, well-managed irrigated field with proper variety choice, timely sowing, balanced fertilizer, and adequate irrigation, yields of 35 to 45 maunds per acre are considered a solid target, while top-performing progressive farms can reach 50 to 60 maunds per acre or more.

National wheat production has fluctuated in recent seasons: Pakistan achieved a record output of around 31.4 to 31.6 million tonnes in 2024-25, but the following 2025-26 season saw a notable decline due to reduced sowing area and dry weather affecting irrigation water availability, illustrating how sensitive wheat output is to both area planted and seasonal weather.

Consumption is growing steadily, driven mainly by population growth, with domestic demand estimated at around 31 to 32 million tonnes per year, meaning Pakistan sometimes needs to import wheat when domestic production falls short of consumption needs. Closing this gap depends heavily on farmers adopting recommended varieties, timely sowing, balanced fertilization, and efficient irrigation, since the yield gap between an average field and a well-managed field remains large and is the biggest untapped source of additional national production.

22. Government Recommendations

Every year before the Rabi sowing season, the Federal Committee on Agriculture (FCA), along with provincial agriculture departments, sets a national wheat production and area target and issues an advisory package covering recommended varieties, sowing windows, and fertilizer doses tailored to each agro-ecological zone. Farmers are encouraged to follow these advisories, available through provincial extension offices, agriculture helplines, and websites of bodies like the Pakistan Agricultural Research Council (PARC) and provincial agriculture departments.

The government also announces a minimum guaranteed support price for wheat each season, meant to protect farmers from price crashes at harvest time, along with procurement operations by PASSCO and provincial food departments to buy wheat directly from farmers at the announced rate. In recent seasons, policy has been shifting toward greater market deregulation, partly linked to broader economic reform commitments, which has made the timely and clear announcement of support prices even more important for farmer planning.

Extension departments in each province run farmer field schools, distribute subsidized or certified seed of new varieties, and promote practices like laser land levelling, zero tillage, balanced fertilizer use based on soil testing, and integrated pest management through provincial agriculture extension wings, the Punjab and Sindh Seed Corporations, and agricultural universities such as the University of Agriculture Faisalabad and Sindh Agriculture University Tandojam.

23. Modern Farming Practices

A range of modern practices is helping Pakistani wheat farmers close the gap between average and potential yields. Laser land levelling is now widely used in Punjab and Sindh to create perfectly flat fields, improving water use efficiency and germination uniformity. Zero tillage and happy-seeder technology allow wheat to be drilled directly into rice or cotton stubble without conventional ploughing, saving time, diesel, and soil moisture in the tightly scheduled rice-wheat and cotton-wheat cropping systems.

Bed-and-furrow (ridge) planting is being promoted as an alternative to flat sowing because it reduces water use, improves drainage, and makes later operations like weeding and top-dressing easier, while also allowing some intercropping options. Mechanical combine harvesters have largely replaced manual harvesting on medium and large farms, reducing post-harvest losses and labour costs, and mobile-based advisory services and SMS alerts are increasingly used by provincial extension departments to send farmers timely reminders about sowing windows, pest outbreaks, and weather warnings.

Use of certified, disease-resistant seed of newly released, higher-yielding varieties, combined with balanced fertilizer application guided by soil testing rather than fixed traditional doses, remains one of the most impactful and accessible modern practices available to farmers of all farm sizes, from smallholders to large commercial operations.

24. Precision Agriculture

Precision agriculture is beginning to take root in Pakistan's wheat belt, especially on larger, more commercially oriented farms in Punjab and Sindh. Satellite and drone-based remote sensing is increasingly used to monitor crop vigour (through vegetation indices like NDVI), detect nutrient stress or disease outbreaks early, and estimate yield before harvest across large areas, supporting both individual farm decisions and provincial crop forecasting.

Variable-rate fertilizer application, guided by soil testing and sometimes by satellite or drone imagery, allows farmers to apply nitrogen, phosphorus, and potassium only where and when the crop actually needs them, rather than a single uniform rate across a whole field, reducing waste and improving nutrient use efficiency. Soil moisture sensors and simple weather-based irrigation scheduling tools are also being piloted through donor-funded and provincial projects to help farmers, especially those relying on expensive tube-well water, decide exactly when to irrigate rather than following a fixed calendar.

Mobile phone apps developed by provincial agriculture departments and research organizations now provide farmers with variety recommendations, pest and disease identification support, and weather-linked sowing and spray advisories, effectively bringing basic precision agriculture tools to smallholders who cannot afford dedicated sensors or drones. While full-scale precision agriculture, with automated machinery and integrated farm management software, is still mostly limited to progressive commercial farms, the building blocks, better data, mobile connectivity, and targeted advisories, are spreading steadily across the wheat-growing provinces.

25. Sustainable Farming

Sustainable wheat farming in Pakistan focuses on maintaining soil health, using water efficiently, and reducing dependence on excessive chemical inputs, all of which are becoming more urgent as groundwater levels fall and reservoirs like Tarbela and Mangla face reduced storage capacity due to silting. Practices such as laser land levelling, bed-and-furrow sowing, and zero tillage all contribute directly to water savings, which is critical since roughly two-thirds of Pakistan's irrigation water depends on snow and glacier melt that is becoming less predictable with climate change.

Maintaining soil organic matter through farmyard manure, compost, green manuring, and proper management of crop residues, rather than burning rice or wheat straw, helps rebuild soil structure and fertility over time and reduces air pollution that burning residue causes in the plains during the sowing season. Crop rotation with legumes, oilseeds, or fodder crops, rather than continuous wheat-rice or wheat-cotton cycles every year, can help break pest and disease cycles and improve soil nitrogen naturally.

Integrated Pest Management, balanced and soil-test-based fertilizer use instead of routine over-application, and the adoption of climate-resilient, drought- and heat-tolerant varieties released by national research institutes are all part of a longer-term strategy to keep wheat production stable as temperatures rise and water becomes scarcer. Government and research institutions increasingly frame these practices not as optional extras but as necessary adjustments for Pakistan's wheat sector to remain food-secure and profitable for farmers in the coming decades.